

17 α -Hydroxyprogesterone induced ovarian growth and vitellogenesis in the freshwater rice field crab *Oziotelphusa senex senex*

P. Ramachandra Reddy, P. Kiranmayi, K. Thanuja Kumari, P. Sreenivasula Reddy *

Department of Biotechnology, Sri Venkateswara University, Tirupati 517 502, India

Received 17 August 2005; received in revised form 22 November 2005; accepted 23 November 2005

Abstract

Vitellogenin, a female-specific phospholipoglycoprotein was used as one of the markers for detecting the induction of reproduction caused by exposure to exogenous chemicals. In the present study, we demonstrated the effect of 17 α -hydroxyprogesterone on the induction of reproduction in the female crab, *Oziotelphusa senex senex*. The crabs were injected with 17 α -hydroxyprogesterone at the doses of 10⁻⁸ or 10⁻⁷ mol/crab on 1st, 7th, 14th, 21st and 28th day of experiment. The mean oocyte diameter and mean ovarian indices in experimental females were increased significantly ($p < 0.001$) when compared with the initial controls. The ovarian vitellogenin content in the experimental females was also increased significantly ($p < 0.001$) in a dose-dependent manner when compared with the initial controls. The vitellogenin was purified and confirmed by double immunodiffusion with anti-vitellin–antisera raised against vitellogenin for this crab. From the results, it is clear that the 17 α -hydroxyprogesterone induces ovarian growth and ovarian vitellogenin synthesis in the crab *O. senex senex*.

© 2005 Elsevier B.V. All rights reserved.

Keywords: Hydroxyprogesterone; Vitellogenin; Oocyte diameter; Ovarian index; Crab

1. Introduction

Vitellogenesis is a process by which female crustaceans produce and sequester nutrients into developing oocytes. In female crustaceans, this process is regulated by vitellogenin inhibiting hormone (VIH) and methyl farnesoate (MF), the major endogenous endocrine hormones (Charniaux-Cotton, 1985; Chang et al., 1993; Reddy and Ramamurthi, 1999; Tsukimura,

2001; Wilder et al., 2002). The VIH inhibits the vitellogenin synthesis, which in turn inhibits the reproduction (Reddy and Ramamurthi, 1999; Tsukimura, 2001; Wilder et al., 2002). Increasing titers of MF in circulation stimulate the hepatopancreas to produce vitellogenin (VTG), the major precursor to egg yolk proteins. This protein is then secreted from the hepatopancreas and transported through the hemolymph to the ovary, where it enters into growing oocytes via specific receptor mediated endocytosis (Laverdure and Soyez, 1988; Jugan and Van Herp, 1989; Warriar and Subramoniam, 2002). Once inside the oocyte, VTG is cleaved into the smaller yolk proteins, by a host of hydrolytic enzymes and finally accumulate in yolk

* Corresponding author. Tel.: +91 877 2249320; fax: +91 877 2249611.

E-mail address: redddy_1955@yahoo.co.in
(P. Sreenivasula Reddy).

globules or granules (Wilder et al., 2002). VTG is characterized as a large molecule of phospholipoglycoprotein and its levels in hemolymph were used as indicator of female reproduction in crustaceans (Wilder et al., 2002).

Several steroids viz., pregnenolone, progesterone, 17 α -hydroxyprogesterone, 17 α -hydroxy pregnenolone, 6 β -hydroxyprogesterone, 20 α -hydroxyprogesterone, 17 β -estradiol, estrone, testosterone, etc., have been identified in different crustaceans (Fingerman et al., 1993; Subramoniam, 2000; Tsukimura, 2001; Wilder et al., 2002; Oetken et al., 2004). It is well known that the circulating steroid hormones induce oocyte growth in oviparous vertebrates. The effect of exogenous steroids on reproduction was demonstrated in different crustacean species but the results were contradictory (Bomirski and Klek-Kawinska, 1976; Souty and Picaud, 1984; Yano, 1987; Yano and Itakura, 1998). The 17 α -hydroxyprogesterone induced vitellogenesis was demonstrated in the kuruma prawn *Penaeus japonicus* (Yano, 1987). Gosh and Ray (1993) reported the 17 β -estradiol stimulated biosynthesis of 17 β -hydroxysteroid dehydrogenase in the fresh water prawn *Macrobrachium rosenbergii*. In the same species, estrogen induced lipogenic activity in the ovary was also reported (Gosh and Ray, 1994). Progesterone induced ovarian maturation was studied in the marine penaeid shrimp *Parapenaeopsis hardwickii* (Kulakarni et al., 1979). Yano and Itakura (1998) demonstrated the effect of 17 β -estradiol on vitellogenesis by ovarian slices in vitro. Recently, Rodriguez et al. (2002) studied the role of 17 α -hydroxyprogesterone and 17 β -estradiol on ovary of the red swamp crayfish *Procambarus clarkii*. Fluctuating levels of steroids, such as 17 β -estradiol and progesterone in the ovary and hemolymph at different reproductive stages of the crab *Scylla serrata* were also reported (Warrier et al., 2001). A significant positive correlation of circulatory estradiol-17 β and progesterone with ovarian development was reported in the giant tiger shrimp *Penaeus monodon* (Quinitio et al., 1994); however, relatively negative relationship between ovarian development and the circulatory levels of steroid hormones was detected in the kuruma prawn *Marsupenaeus japonicus* (Summavielle et al., 1995; Okumura and Sakiyama, 2004).

In several crustaceans, increased oocyte diameter and change in color of the ovary were used as criteria to determine ovarian maturation (Charniaux-Cotton and Payen, 1988). The ovarian maturation was also determined by considering the concentration of vitellogenin in the hemolymph and accumulation of yolk globules in oocytes (Tsukimura, 2001). In view of

this, the present study was aimed to elucidate the role of 17 α -hydroxyprogesterone on female reproduction in the fresh water crab, *Oziotelphusa senex senex*, using ovarian growth and ovarian vitellogenin levels as parameters.

2. Material and methods

2.1. Maintenance of animals

Adult intermolt stage C₄ (determined by observing the setae in the mastigobranch of third maxillipede as described by Reddy, 1991) female crabs (*O. senex senex*) with a body weight of 32 $\pm$ 2 g and body size of 40 $\pm$ 5 mm were collected from the rice fields located in and around Tirupati. The animals were maintained under controlled conditions (12-h light/12-h dark; temperature 27 $\pm$ 1 °C; humidity 75%) in large glass aquaria with sufficient ambient medium (salinity: 0.5 ppt). They were acclimatized for at least 10 days before being used for experimentation. The crabs were fed daily with sheep meat ad libitum and the water was changed 3 h after feeding.

2.2. Test hormone

Test hormone, 17 α -hydroxyprogesterone (Trade name: Proluton depot) selected for the study was manufactured by German Remedies, Goa and purchased from local drug store, India. The 17 α -hydroxyprogesterone is an oil based injection containing castor oil IP and benzyl benzoate IP (1:1.7).

2.3. Experimentation

One hundred crabs were selected and divided into 4 equal groups. The crabs in the first group served as controls, did not receive any treatment and were sacrificed on day 1 of experiment. The animals in the second group served as concurrent control and was treated same as the experimental groups, but received 10 μ l of solvent (castor oil/benzyl benzoate; 1:1.7). The crabs in the third group and fourth group were injected with 17 α -hydroxyprogesterone 10 $^{-8}$ or 10 $^{-7}$ mol/crab, respectively, in 10 μ l volume on 1st, 7th, 14th, 21st and 28th day of experiment. The crabs were sacrificed with ice anesthetization on day 35 of the experiment. The doses of 17 α -hydroxyprogesterone used in the present study were similar to those employed in the previous studies with several crustacean species (Nagabhushanam et al., 1980, 1983; Rodriguez et al., 2002).

2.4. Measurement of gonad index

After sacrifice, crabs from each group were weighed and the ovaries were excised, cleaned in crustacean saline, weighed and fixed in aqueous Bouin's fluid. The ovarian indices were measured using the following standard formula:

Wet weight of the ovary/Weight of the crab $\times 100$

2.5. Histological studies of ovary

The ovarian sections were prepared according to the method described by Brancraft and Stevens (1982). The ovaries were fixed in Bouin's fluid (picric acid/formaldehyde/acetic acid; 75:25:5) for 24 h immediately after excise and dehydrated with ascending alcoholic series, cleared in xylene and then embedded in paraffin wax (m.p. 56–58 °C). The sections were made in 7 μ m thickness, stained with Hematoxylin followed by counter staining in alcoholic Eosin.

The diameter of 25 oocytes from each ovary was measured using an ocular micrometer under compound microscope (Olympus). The ovaries of the remaining crabs from each group were used for vitellogenin isolation. The characteristic (morphological and histological) features of the ovary in the crab *O. senex senex* were described (Table 1).

2.6. Isolation of vitellogenin

Vitellogenin was isolated from the ovaries of crab *O. senex senex* by the method described by Tsukimura et al. (2000). In brief, the ovarian tissues (10% w/v) were homogenized in homogenization buffer (containing 0.1M NaCl, 0.05M Tris, 1 mM ethylene diamine tetra-acetic acid and 0.1% Tween 20 with 10 mg/ml PMSF; pH 7.8) using an ice-cold glass–glass homogenizer. The homogenate was centrifuged at 4000 $\times$ g for 5 min at 4 °C. The resultant supernatant was again centrifuged at 20,000 $\times$ g for 20 min at 4 °C. To the supernatant, saturated ammonium sulphate (SAS) was added to produce 25% SAS solution. After incubation for 1 h at 4 °C, the solution was centrifuged at 20,000 $\times$ g for 10 min at 4 °C. The supernatant was collected and SAS was added to produce 40%, 50% and 60% SAS solution sequentially. The pellets of 60% SAS solution was suspended in appropriate volume of homogenization buffer and dialyzed thrice at 4 °C for 12 h each against homogenization buffer. The purified vitellogenin was stored at –20 °C for further use.

Table 1

Characteristic features of different reproductive stages in the crab *Ozotellphusa senex senex*

Reproductive stage	Characteristic features
(a) Immature	(i) Ovary is translucent to white in color. (ii) Thick ovarian wall with a centrally located germanium surrounded by a number of oocytes (iii) Diameter of oocytes ranges between 15 and 23 μ m
(b) Vitellogenesis	
Vitellogenic stage I	(i) The color of the ovary is light yellow (ii) Yolk globules are deposited at the periphery with prominent nucleus and nucleolus in the oocytes (iii) Oocytes are surrounded by follicle cells (iv) Diameter of oocytes lies between 30 and 40 μ m
Vitellogenic stage II	(i) Ovary is orange in color (ii) Peripherally located yolk globules move towards the centre, gradually replacing the protein yolk (iii) Diameter of oocytes lies between 50 and 60 μ m
Vitellogenic stage III	(i) Ovary color is dark brown or bright orange (ii) Large accumulation of yolk globules in oocytes (iii) Oocyte diameter reaches maximum 65–80 μ m
(c) Spent ovary	(i) Ovary is very thin and consists of several active follicles

2.7. Preparation of antiserum against vitellogenin

Polyvalent antiserum against the purified vitellogenin was raised in a rabbit. An emulsion of the purified vitellogenin fraction and Freund's complete (first injection) or Freund's incomplete adjuvant (subsequent injections) was injected intra-dermally into the back of the rabbit. Rabbits were immunized weekly and after six injections were test bled. The resulting antisera was serially diluted and reacted with purified vitellogenin using double diffusion technique. Once a sufficient titer was reached, rabbits were exsanguinated and the polyclonal antiserum was collected and stored at –80 °C.

2.8. Electrophoresis

The purity of the isolated vitellogenin was checked by sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE) according to the method of Laemmli (1970) in the presence of 0.1% SDS on a vertical slab gel system. The gels were stained with Coomassie brilliant blue and destained in ethanol (20%) and acetic acid (7%) solution. Molecular weights of the vitellogenin subunits were calculated

graphically using Genei medium range molecular weight markers.

2.9. Double immunodiffusion

Double immunodiffusion experiments were performed using 0.5% agarose gels according to the published procedure of Ouchterlony (1958) using the antibodies raised against vitellogenin for this crab, to confirm the vitellogenin protein in the purified samples. After loading the samples, the plates were placed in a moist chamber for 12 h. The plates were blotted with a thick stack of filter papers, stained with Coomassie brilliant blue for 2 h and destained in a solution containing methanol, acetic acid and water (20:7:73). The intensity of the precipitin bands was analyzed using image analyzer (Chemi Imager™ Ready, USA).

2.10. Statistical analysis

Data are expressed as mean±S.D. Results were analyzed using SPSS 10.0 version.

3. Results

The mean oocyte diameter of control crabs was 19.2 µm (Fig. 1). At the end of the experiment (concurrent controls), the diameter of the oocytes averaged 19.2 µm, which were not significantly different from the control values, indicating that no significant ovarian growth had occurred during the experimental period. For the crabs that received the smallest dose of 17α-hydroxyprogesterone (10^{-8} mol/crab), the oocyte diameters were significantly ($p<0.0001$) larger than the corresponding values of the control group. With increas-

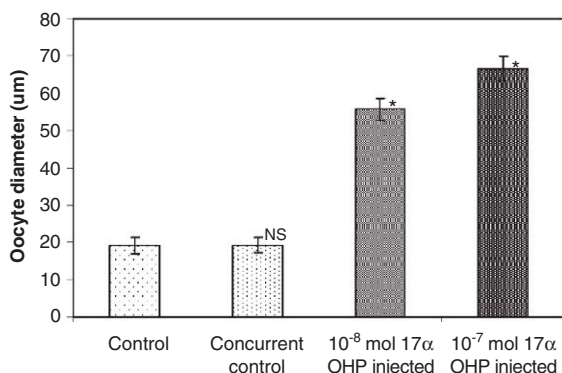


Fig. 1. The effect of injection of 17α-hydroxyprogesterone (17α-OHP) on oocyte diameter in the crab *O. senex senex*. Asterisks denote values are significantly ($p<0.0001$) different from control. NS=not significant.

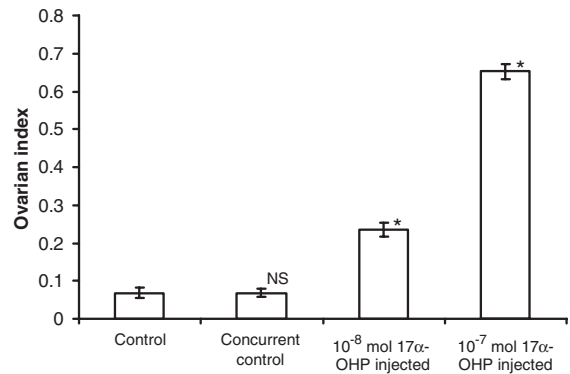


Fig. 2. The effect of injection of 17α-hydroxyprogesterone (17α-OHP) on ovarian indices in the crab *O. senex senex*. Asterisks denote values are significantly ($p<0.0001$) different from control. NS=not significant.

ing dose of 17α-hydroxyprogesterone (10^{-7} mol/crab), the oocyte diameters further increased and the mean oocyte diameter reached 66.6 µm (Fig. 1).

The color of the ovaries was white and the average ovarian index was 0.0685% for the control crabs and 0.0682% for the concurrent control crabs (Fig. 2). The ovaries from the 17α-hydroxyprogesterone injected groups were significantly ($p<0.0001$) larger and showed a dose–response; the 10^{-8} mol 17α-hydroxyprogesterone group had orange red colored ovaries and the mean ovarian index was 0.235%, whereas the group received the higher dose of 10^{-7} mol 17α-hydroxyprogesterone had dark brown colored ovaries which were significantly larger than controls having a mean ovarian index of 0.652% (Fig. 2).

The purified vitellogenin isolated from vitellogenic ovaries was separated into 3 bands of 74.3, 83.3 and 100 kDa in SDS-PAGE (Fig. 3). The ovarian vitellogenin content increased significantly ($p<0.0001$) in 17α-

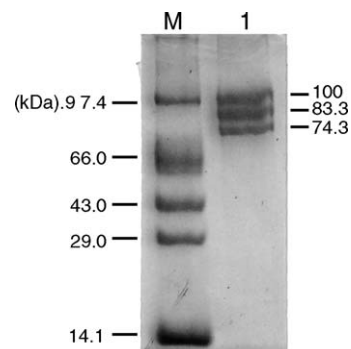


Fig. 3. The SDS electrophoresis pattern of crab ovary vitellogenins. M: molecular weight markers. Lane 1: purified vitellogenins from ovary.

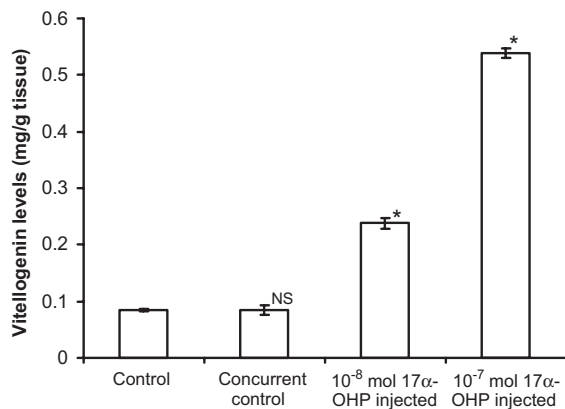


Fig. 4. The effect of injection of 17 α -hydroxyprogesterone (17 α -OHP) on ovarian vitellogenin levels in the crab *O. senex senex*. Asterisks denote values are significantly ($p < 0.0001$) different from control. NS=not significant.

hydroxyprogesterone injected crabs when compared with the initial control crabs (Fig. 4) and showed dose–response. The purified vitellogenin isolated from the ovaries of crabs injected with 17 α -hydroxyprogesterone showed clear precipitin bands (Fig. 5). Whereas no precipitin reaction was observed with the samples of initial and concurrent controls (Fig. 5).

The ovary of concurrent control crabs and initial controls had a thick ovarian wall with a centrally located germanium surrounded by a number of oocytes (Fig. 6A). The oocytes of the ovary from 10⁻⁸ mol 17 α -hydroxyprogesterone/crab injected females appeared to be filled with protein yolk suggesting that the oocyte entered in to the previtellogenic stage (Fig. 6B). A still more advanced stage (vitellogenesis stage II) of oocytes was seen in 10⁻⁷ mol 17 α -hydroxyprogesterone/crab

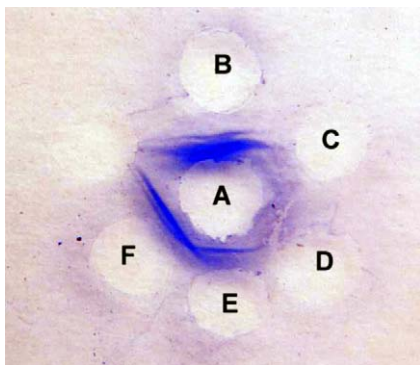


Fig. 5. Immunodiffusion of vitellogenin isolated from the ovaries of initial control, concurrent control and 17 α -hydroxyprogesterone injected crabs. (A) Vitellogenin antibody; (B) purified vitellogenin; (C) initial control; (D) concurrent control; (E) 10⁻⁸ mol 17 α -hydroxyprogesterone/crab; (F) 10⁻⁷ mol 17 α -hydroxyprogesterone/crab.

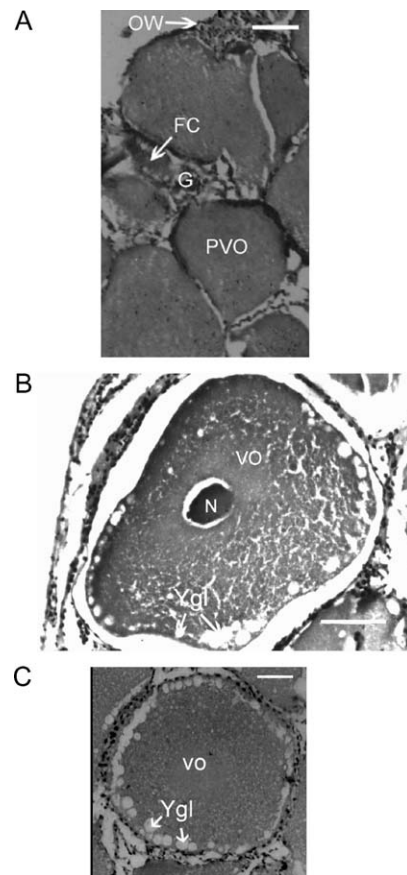


Fig. 6. The effect of injection of 17 α -hydroxyprogesterone on ovarian histology in the crab *O. senex senex*. Scale: horizontal bar=15 μ m. (A) Transverse section of immature ovary from concurrent control females showing pre-vitellogenic oocytes (PVO) with thick ovarian wall (OW), germanium (G) and follicle cells (FC). (B) Transverse section of the ovary from 10⁻⁸ mol hydroxyprogesterone/crab injected female crabs showing yolk globules (Ygl) deposited at the periphery with prominent nucleus (N) and nucleolus. The vitellogenic oocyte (VO) was surrounded by follicle cells (FC). (C) Transverse section of the ovary from 10⁻⁷ mol hydroxyprogesterone/crab injected female crabs showing peripherally located yolk globules (Ygl) move towards the centre, gradually replacing the protein yolk in the vitellogenic oocyte (VO).

injected crabs. The peripherally located yolk globules tend to move towards centre, gradually replacing the protein yolk (Fig. 6C).

4. Discussion

Several methods were employed to study the reproductive status of decapod crustaceans. Jegla (1966) used recording of ovigerous females as an index to determine the reproductive cycle in cave crayfish *Orconectes pel-lucidus inermis*. Use of gonad index and the change of color of gonad were employed for identification of

breeding period in crustaceans (Haley, 1972; Chandran, 1968). Histological observation of gonad was employed as a sensitive method of assessing reproductive status of crustaceans (Zerbib, 1977). The diameter of the oocyte was also used as a tool to determine the condition of reproduction. Recently, several workers used the levels of vitellogenin in gonad as a marker for induction of reproduction in crustaceans (Tsukimura, 2001; Wilder et al., 2002).

In the present study, vitellogenin of *O. senex senex* was purified from vitellogenic stage II ovaries and its molecular mass was estimated as 74.3, 83.3 and 100 kDa by SDS-PAGE. The vitellogenin subunit composition of *Pandalus hypsinotus* (86, 98 and 190 kDa, Okumura et al., 2004), *P. kessleri* (81 and 110 kDa, Qunitio et al., 1989), *M. rosenbergii* (90, 102 and 199 kDa, Wilder et al., 1994), and in *M. nipponense* (90 and 102 kDa, Han et al., 1994) were reported earlier. Where as electrophoresis of purified vitellogenin isolated from mysid shrimp (*Neomysis integer*) by SDS-PAGE resulted in 6 subunits with molecular masses of 51, 55, 62, 66, 84 and 89 kDa (Ghekiere et al., 2004). Tuberty et al. (2002) isolated vitellogenin from five different estuarine crustacean species, and depending on the species, these authors detected 6–12 subunits. There appears to be considerable variability in the subunit composition of vitellogenin in crustacean species.

The changes in ovarian index, oocyte diameter, histological alterations and vitellogenin levels in ovaries during 17 α -hydroxyprogesterone treatment were also determined in the present study. The increased oocyte diameter and ovarian indices with increased ovarian vitellogenin in crabs injected with 17 α -hydroxyprogesterone indicates the role of 17 α -hydroxyprogesterone in inducing reproduction in the crab.

Literature on the endocrinology of crustaceans suggests that estrogens are stimulating hormones for female sexual maturation and egg laying (Andersen et al., 1999). Vertebrate-like steroids have been shown to induce significant oocyte growth in several crustaceans (Nagabhushanam et al., 1980, 1983; Yano, 1985). Yano (1987) observed 9.6-fold increase of vitellogenin in the female prawn *Penaeus japonicus* injected with 17 α -hydroxyprogesterone. The 17 β -estradiol induced vitellogenesis in the ovarian pieces under in vitro conditions was also reported in the same prawn (Yano and Itakura, 1998). A positive correlation between circulatory steroids and vitellogenin levels has been reported for shrimp (Qunitio et al., 1994) and crabs (Shih, 1997) but correlation between increased levels of steroid hormones and vitellogenesis is not observed in kuruma

prawn (Okumura and Sakiyama, 2004). Moreover, inhibition of reproduction was observed in red swamp crayfish, *P. clarkii* by 17 α -hydroxyprogesterone and 17 β -estradiol (Rodriguez et al., 2002). The reason for the difference is not known, but it may be related to possible differences in species and animal conditions. Further studies are required to examine the species difference in the roles of vertebrate-type steroid hormones.

Recently, Billingham et al. (2000) demonstrated that exposure to the natural estrogen 17 β -estradiol and a xenoestrogen 4-nonylphenol to barnacle larvae resulted in the expression of a larvae storage protein called 'cypris major protein', which is structurally and immunologically related to barnacle vitellogenin. This suggests that 17 α -hydroxyprogesterone may play an important role in the ovarian development of *O. senex senex*. Because 17 α -hydroxyprogesterone stimulated ovarian growth and maturation in *O. senex senex*, it may be possible to use this hormone to stimulate reproduction in commercially important crustaceans. Supplementing the diet with steroid hormones may accelerate reproductive activity and this might be used as a tool to facilitate crustacean aquaculture. Studies are underway in our laboratory to test the effectiveness of 17 α -hydroxyprogesterone in inducing ovarian development and spawning in brood-stock females of aquaculturally important crustacean species.

Acknowledgements

This work was supported by a grant (SP/SO/C42/2001) from Department of Science and Technology, New Delhi to PSR. We also thank Prof. K.V.S. Sarma, Department of Statistics, S.V. University, Tirupati for statistical analysis of data and Mr. S. Uma-sankar for maintaining the animals in the laboratory. Part of the data was submitted to Sri Venkateswara University in partial fulfillment for the award of the Degree of Master of Science in Biotechnology by PK and KTK.

References

- Andersen, H.R., Halling-Sorensen, B., Kusk, O.K., 1999. A parameter for detecting estrogenic exposure in the copepod *Acartia tonsa*. *Ecotoxicol. Environ. Saf.* 44, 56–61.
- Billingham, Z., Clare, A.S., Matsumura, K., Depledge, M.H., 2000. Induction of cypris major protein in barnacle larvae by exposure to 4-n-nonylphenol and 17 β -oestradiol. *Aquat. Toxicol.* 47, 203–212.
- Bomirski, A., Klek-Kawinska, E., 1976. Stimulation of oogenesis in the sand crab, *Crangon crangon* by human gonadotropin. *Gen. Comp. Endocrinol.* 30, 239–242.

- Brancraft, J.D., Stevens, A., 1982. Theory and Practice of Histological Techniques, 2nd edn. Churchill Livingstone, New York.
- Chandran, M.R., 1968. Studies on the crab, *Charybdis variegata*: I. Reproduction and maturation cycle in relation to breeding periodicities. *Proc. Indian Acad. Sci.* 67, 215–223.
- Chang, E.S., Bruce, M.J., Tamone, S.L., 1993. Regulation of crustacean molting: a multi-hormonal system. *Am. Zool.* 33, 324–329.
- Charniaux-Cotton, H., 1985. Vitellogenesis and its control in malacostracan Crustacea. *Am. Zool.* 25, 197–206.
- Charniaux-Cotton, H., Payen, G.G., 1988. Crustacean reproduction. In: Laufer, H., Downer, R.G.H. (Eds.), *Endocrinology of Selected Invertebrate Types*. Aln. Liss. Inc., New York, pp. 279–303.
- Fingerman, M., Nagabhushanam, R., Sarojini, R., 1993. Vertebrate-type hormones in crustaceans: localization, identification and functional significance. *Zool. Sci.* 10, 13–29.
- Ghekiere, A.N., Verslycke, T., De Smet, L., Van Beeumen, J., Janssen, C.R., 2004. Purification and characterization of vitellin from the estuarine mysid *Neomysis integer* (Crustacea; Mysidacea). *Comp. Biochem. Physiol.* 138A, 427–433.
- Gosh, D., Ray, A.K., 1993. 17 β -Hydroxysteroid dehydrogenase activity of ovary and hepatopancreas of freshwater prawn *Macrobrachium rosenbergii*: relation to ovarian condition and estrogen treatment. *Gen. Comp. Endocrinol.* 89, 248–254.
- Gosh, D., Ray, A.K., 1994. Estrogen stimulated lipogenic activity in the ovary of the freshwater prawn *Macrobrachium rosenbergii*. *Invertebr. Reprod. Dev.* 25, 43–47.
- Haley, S.R., 1972. Reproductive cycling in the ghost crab, *Ocypode quadrata* (Fabricius) (Brachyura; Ocypodidae). *Crustaceana* 23, 1–11.
- Han, C.H., Okumura, T., Suzuki, Y., Aida, K., Hanyu, I., 1994. Immunocytochemical identification of the site of vitellogenin synthesis in the freshwater prawn *Macrobrachium nipponense*. *Fish. Sci.* 60, 149–154.
- Jegla, T.C., 1966. Reproductive and molting cycles in cave crayfish. *Biol. Bull.* 130, 345–358.
- Jugan, P., Van Herp, F., 1989. Introductory study of an oocyte membrane protein that specifically binds vitellogenin in the crayfish, *Orconectus limosus*. *Invertebr. Reprod. Dev.* 16, 149–154.
- Kulakarni, G.K., Nagabhushanam, R., Joshi, P.K., 1979. Effect of progesterone on ovarian maturation in a marine penaeid prawn *Parapenaeopsis hardwickii*. *Indian J. Exp. Biol.* 17, 986–987.
- Laemmli, U.K., 1970. Cleavage of structural proteins during the assembly of bacteriophage T₄. *Nature* 227, 680–685.
- Laverdure, A.M., Soye, D., 1988. Vitellogenin receptor from lobster oocyte membrane: solubilization and characterization by a solid phase binding assay. *Int. J. Invertebr. Reprod. Dev.* 13, 251–266.
- Nagabhushanam, R., Joshi, P.K., Kulkarni, G.K., 1980. Induced spawning in prawn *Parapenaeopsis styliifera* (H. Milne Edwards) using a steroid hormone 17hydroxy-progesterone. *Indian J. Mar. Sci.* 9, 227.
- Nagabhushanam, R., Kulkarni, G.K., Joshi, P.K., 1983. Effect of some vertebrate steroid hormones on the ovarian maturation and spawning in Indian penaeid prawns. *Marathwada Univ. J. Sci.* 22, 89–98.
- Oetken, M., Bachmann, J., Schulte-Oehlmann, U., Oehlmann, J., 2004. Evidence for endocrine disruption in invertebrates. *Int. Rev. Cyt.* 236, 1–44.
- Okumura, T., Sakiyama, K., 2004. Hemolymph levels of vertebrate-type steroid hormones in female kuruma prawn *Marsupenaeus japonicus* (Crustacea: Decapoda: Penaeida) during natural reproductive cycle and induced ovarian development by eyestalk ablation. *Fish. Sci.* 70, 372–380.
- Okumura, T., Yoshida, K., Nikaido, H., 2004. Ovarian development and hemolymph vitellogenin levels in laboratory-maintained protandric shrimp, *Pandalus hypsinotus*: measurement by a newly developed time-resolved fluoroimmunoassay (TR-FIA). *Zool. Sci.* 21, 1037–1047.
- Ouchterlony, O., 1958. Diffusion-in-gel methods for immunological analysis. *Prog. Allergy* 5, 1–78.
- Qunitio, E.T., Hara, A., Yamauchi, K., Mizushima, T., Fuji, A., 1989. Identification and characterization of vitellin in a hermaphrodite shrimp *Pandalus kessleri*. *Comp. Biochem. Physiol.* 94B, 445–451.
- Qunitio, E.T., Hara, A., Yamauchi, K., Nakao, S., 1994. Changes in the steroid hormone and vitellogenin levels during the gametogenic cycle of the giant tiger shrimp, *Penaeus monodon*. *Comp. Biochem. Physiol.* 109C, 21–26.
- Reddy, P.S., 1991. Neuroendocrine control of molting in freshwater field crab *Oziotelphusa senex senex* Fabricius. *Trop. Freshw. Biol.* 2, 213–222.
- Reddy, P.S., Ramamurthi, R., 1999. Recent trends in crustacean endocrine research. *PINSA* 65, 15–32.
- Rodriguez, E.M., Lopez-Greck, L.S., Medesani, D.A., Laufer, H., Fingerman, M., 2002. Effects of methyl farnesoate, alone and in combination with other hormones, on ovarian growth of the red swamp crayfish *Procambarus clarkii*, during vitellogenesis. *Gen. Comp. Endocrinol.* 125, 34–40.
- Shih, J.T., 1997. Sex steroid-like substances in the ovaries, hepatopancreas, and body fluid of female *Mictyris brevidactylus*. *Zool. Stud.* 36, 136–145.
- Souty, C., Picaud, J.L., 1984. Effet de l'injection d'une gonadotropine numain sur la synthese et la liberation de la vitellogenine par le tissu adipeux du crustace isopode marine *Idotea baltica basteri* Adouin. *Gen. Comp. Endocrinol.* 54, 418–421.
- Subramoniam, T., 2000. Crustacean ecdysteroids in reproduction and embryogenesis. *Comp. Biochem. Physiol.* 125C, 135–156.
- Summavielle, T., Silva, L., Reis-Henriques, M.A., Baldaia, L., Coimbra, J., 1995. Haemolymph unconjugated and conjugated steroids during reproduction in *Penaeus japonicus* (Crustacea: Decapoda). *Neth. J. Zool.* 45, 64–67.
- Tsukimura, B., 2001. Crustacean vitellogenesis: its role in oocyte development. *Am. Zool.* 41, 465–476.
- Tsukimura, B., Bender, J.S., Linder, C.J., 2000. Development of an anti-vitellogenin ELISA for the assessment of reproduction in the ridgeback shrimp, *Sicyonia ingentis*. *Comp. Biochem. Physiol.* 127A, 215–224.
- Tuberty, S.R., Nates, S.F., McKenney Jr., C.L., 2002. Polyclonal antisera against estuarine crustacean vitellins. In: Escobar-Briones, E., Alvarez, F. (Eds.), *Modern Approaches to the Study of Crustacea*. Kluwer Academic/Plenum Publishers, New York, pp. 29–37.
- Warrier, S.R., Subramoniam, T., 2002. Receptor mediated yolk protein uptake in the crab *Scylla serrata*: crustacean vitellogenin receptor recognizes related mammalian serum lipoproteins. *Mol. Reprod. Dev.* 61, 536–548.
- Warrier, S.R., Tirumalai, R., Subramoniam, T., 2001. Occurrence of vertebrate steroids, estradiol-17 β and progesterone in the reproducing females of the mud crab *Scylla serrata*. *Comp. Biochem. Physiol.* 130A, 283–294.
- Wilder, M.N., Okumura, T., Suzuki, Y., Fusetani, N., Aida, K., 1994. Vitellogenin production induced by eyestalk ablation in juvenile giant freshwater prawn *Macrobrachium rosenbergii*

- and trial methyl farnesoate administration. Zool. Sci. 11, 45–53.
- Wilder, M.N., Subramoniam, T., Aida, K., 2002. Progress in vitellogenesis. In: Raikhel, A.S., Sappington, T.W. (Eds.), Reproductive Biology of Invertebrates. Science Publishers Inc., Plymouth, UK, pp. 131–174.
- Yano, I., 1985. Induced ovarian maturation and spawning in greasy-back shrimp, *Metapenaeus ensis*, by progesterone. Aquaculture 47, 223–229.
- Yano, I., 1987. Effect of 17 α -hydroxyprogesterone on vitellogenin secretion in kuruma prawn, *Penaeus japonicus*. Aquaculture 61, 49–57.
- Yano, I., Itakura, N., 1998. Hormonal control of vitellogenin synthesis in the ovary of the kuruma prawn. Proceedings and Abstracts, Fourth International Crustacean Congress, Amsterdam. August, vol. 81, p. 64.
- Zerbib, C., 1977. Protein absorption during oogenesis of the crustacean amphipoda *Orchestia gammarellus* (Pallas) demonstration using horseradish peroxidase. Comp. Rend. 28, 757–760.